

CPEN 416 / EECE 5710

## **Lecture 08: The oracle, query complexity, and Deutsch's algorithm**

Tuesday 7 October 2025

# Announcements

- No quiz today
- Midterms available for pickup after class
- Project details coming soon (start thinking about groups)

# Module 2 learning outcomes

## Learning outcomes:

- explain what it means for an algorithm to have a quantum speedup
- define quantum oracles and query complexity
- implement oracles and Grover's algorithm in PennyLane
- identify the different components of the quantum compilation stack
- define and list common universal gate sets
- estimate the resources required to run a quantum algorithm
- perform simple circuit optimizations in PennyLane

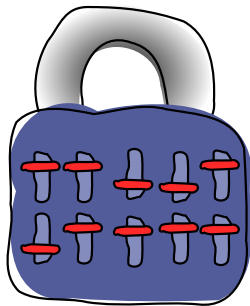
# Today

## Learning outcomes:

- Define the query complexity of an algorithm
- Describe multiple strategies for implementing an *oracle* query with a quantum circuit
- Apply the phase kickback trick
- Implement Deutsch's algorithm to determine whether a Boolean function is constant or balanced (in theory and in PennyLane)

# Oracles: motivating problem

Suppose we would like to find the combination for a “binary” lock:



How do we solve this classically?

Image credit: Codebook node A.1

# Idea: use superposition

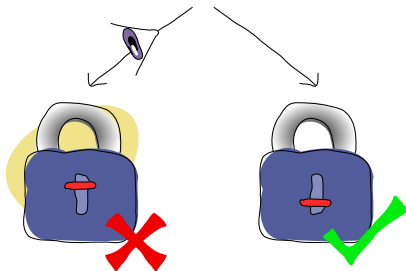
Can we do better with a quantum computer?

Idea: take  $n$  qubits and construct a *uniform* superposition of all combinations

**Exercise:** design a circuit to create this state.

# Idea: use superposition

Measurements are probabilistic - the solution just being “in” the superposition doesn’t make it easier to solve the problem.



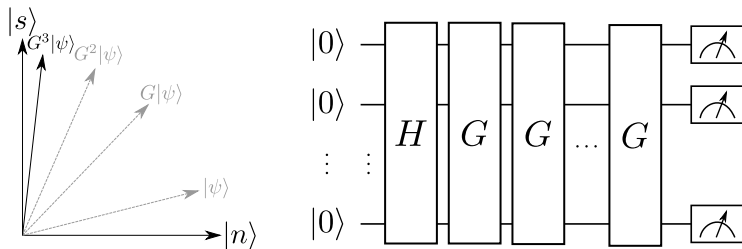
Quantum computers are **NOT** faster because they “compute everything at the same time.”

Image credit: Codebook node A.1

# Solving problems with quantum computers

Can we solve this problem better with a quantum computer?

Yes: **amplitude amplification**, and **Grover's algorithm**



We will explore the algorithmic primitives, and see other cases where quantum computers have a **speed up in query complexity**.



# Oracles

Revisit the lock problem and model a “try” as a function. Let

- $\mathbf{x}$  be an  $n$ -bit string (an input combination)
- $\mathbf{s}$  be the correct combination

Don't care *how*  $f(\mathbf{x})$  is evaluated, only that it gives a “yes/no” answer.  $f(\mathbf{x})$  is a black box, or **oracle**.

Every time we try a combination, we are **querying the oracle**.  
The amount of queries is the **query complexity**.

# Quantum oracles

Must determine

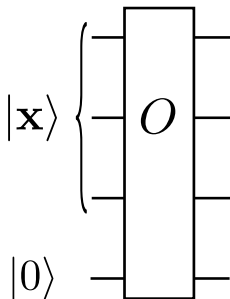
- a mathematical description of an oracle
- where to record the output, i.e., the value of  $f(\mathbf{x})$

Idea 1: encode the result in the state of an additional qubit.

**Exercise:** Consider a 2-qubit system where  $f(01) = 1$ , and  $f(\mathbf{x}) = 0$  for all other  $\mathbf{x}$ . What is the action of the oracle?

| Input state           | Output state |
|-----------------------|--------------|
| $ 00\rangle 0\rangle$ |              |
| $ 01\rangle 0\rangle$ |              |
| $ 10\rangle 0\rangle$ |              |
| $ 11\rangle 0\rangle$ |              |

# Quantum oracles



$$O|000\rangle|0\rangle = |000\rangle|0\rangle$$

$$O|001\rangle|0\rangle = |001\rangle|0\rangle$$

$$O|010\rangle|0\rangle = |010\rangle|0\rangle$$

$$O|011\rangle|0\rangle = |011\rangle|0\rangle$$

$$O|100\rangle|0\rangle = |100\rangle|0\rangle$$

$$O|101\rangle|0\rangle = |101\rangle|0\rangle$$

$$O|110\rangle|0\rangle = |110\rangle|1\rangle$$

$$O|111\rangle|0\rangle = |111\rangle|0\rangle$$

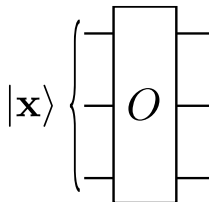
# Quantum oracles

Idea 2: encode the result in the phase

**Exercise:** Consider a 2-qubit system where  $f(11) = 1$ , and  $f(\mathbf{x}) = 0$  for all other  $\mathbf{x}$ . What is the action of the oracle?

| Input state  | Output state |
|--------------|--------------|
| $ 00\rangle$ |              |
| $ 01\rangle$ |              |
| $ 10\rangle$ |              |
| $ 11\rangle$ |              |

# Quantum oracles



$$O|000\rangle = |000\rangle$$

$$O|001\rangle = |001\rangle$$

$$O|010\rangle = |010\rangle$$

$$O|011\rangle = |011\rangle$$

$$O|100\rangle = |100\rangle$$

$$O|101\rangle = |101\rangle$$

$$O|110\rangle = -|110\rangle$$

$$O|111\rangle = |111\rangle$$

# Oracle circuits

**Exercise:** develop two quantum circuit implementations for a 2-input oracle with function  $f(00) = 1$ .

# Oracle circuits

**Exercise:** construct two-qubit oracle circuits (“type 1” oracle) for the following four functions.

| Name  | Action                       | Name  | Action                       |
|-------|------------------------------|-------|------------------------------|
| $f_1$ | $f_1(0) = 0$<br>$f_1(1) = 0$ | $f_2$ | $f_2(0) = 1$<br>$f_2(1) = 1$ |
| $f_3$ | $f_3(0) = 0$<br>$f_3(1) = 1$ | $f_4$ | $f_4(0) = 1$<br>$f_4(1) = 0$ |

# Deutsch's algorithm

**Motivation:** You have access to an oracle that implements one of the following functions:

| Name  | Action       | Name  | Action       |
|-------|--------------|-------|--------------|
| $f_1$ | $f_1(0) = 0$ | $f_2$ | $f_2(0) = 1$ |
|       | $f_1(1) = 0$ |       | $f_2(1) = 1$ |
| $f_3$ | $f_3(0) = 0$ | $f_4$ | $f_4(0) = 1$ |
|       | $f_3(1) = 1$ |       | $f_4(1) = 0$ |

Functions  $f_1$  and  $f_2$  are *constant*,  $f_3$  and  $f_4$  are *balanced*.

How many **classical** oracle queries to determine if the oracle function is constant or balanced? How many **quantum** queries?



# Phase kickback

The secret is the *phase kickback* trick.

**Exercise:** apply  $\text{CNOT}_{01}$  to the following two-qubit states.

$$|0\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right), \quad |1\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

# Deutsch's algorithm

Let  $U_f$  be an oracle for one of the four functions:

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

Initializing the second qubit to  $|-\rangle$  allows us to learn  $f(0) \oplus f(1)$  with a single oracle query.

**Q:** How does  $f(0) \oplus f(1)$  relate to properties of the function?

# Deutsch's algorithm

$$U_f|x\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) =$$
$$=$$

If  $f(x) = 0$ ,

$$U_f|x\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) =$$

If  $f(x) = 1$ ,

$$U_f|x\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) =$$

# Deutsch's algorithm

Remember how we generalized the result for CNOT:

$$CNOT \left( |b\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) = (-1)^b |b\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right),$$

So we can write

# Deutsch's algorithm

How to use this to get  $f(0) \oplus f(1)$ ?

$$U_f \left( \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$=$$
$$=$$
$$=$$
$$=$$

# Deutsch's algorithm

$$U_f \left( \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \frac{|0\rangle + (-1)^{f(0) \oplus f(1)} |1\rangle}{\sqrt{2}} \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

If the function is constant,  $f(0) \oplus f(1) = 0$  and the state is

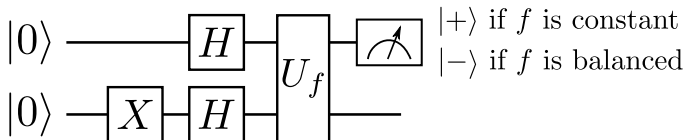
$$U_f \left( \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) =$$

But if the function is balanced,  $f(0) \oplus f(1) = 1$  and the state is

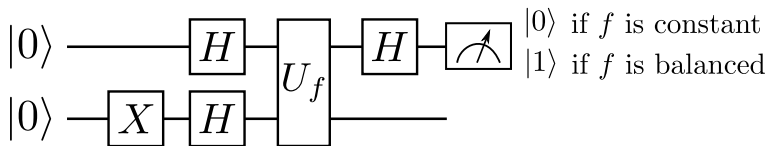
$$U_f \left( \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) =$$

# Deutsch's algorithm

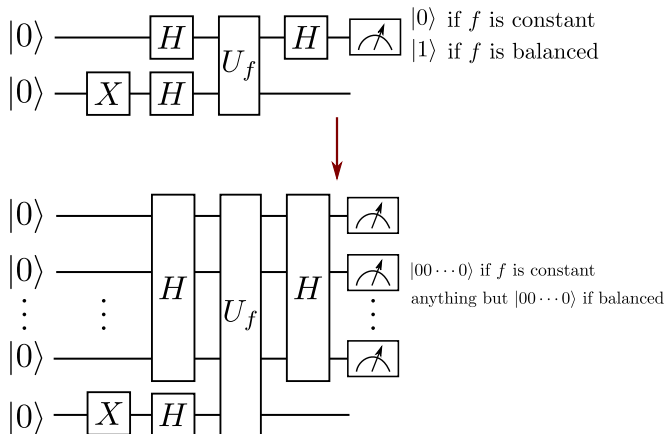
Deutsch's algorithm as a circuit:



Equivalently,



# Generalization: Deutsch-Jozsa algorithm



$2^{n-1} + 1$  classical queries in worst case; still only 1 quantum query.  
(Challenge: try implementing it yourself to check if this works!)



# Oracle-based algorithms

A few other interesting algorithms:

## Bernstein-Vazirani algorithm (practice A2)

Given  $f : \{0, 1\}^n \rightarrow \{0, 1\}$  such that  $f(x) = x \cdot s$  for some secret bitstring  $s$ . Find  $s$  using the fewest number of oracle queries.

## Simon's algorithm

Given  $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$  and promised that for some non-trivial bit string  $s$ ,  $f(x) = f(y)$  iff  $x \oplus y = s$ . Find  $s$  using the fewest number of oracle queries.

# Grover's quantum search algorithm

Let's break that lock!

Input combination is an  $n$ -bit string. Correct combination is  $\mathbf{s}$ .  
How many oracle queries the oracle to find the solution?



Classical:

Quantum:

# Grover's quantum search algorithm

Start with a uniform superposition, then *amplify* the amplitude of the solution state  $|s\rangle$ .

In other words, go from the uniform superposition

to something that looks like:

Q: Why do we want a state of this form?

# Next time

Content:

- Amplitude amplification and Grover's algorithm

Action items:

- 1 Start thinking about project groups

Recommended reading:

- For today: Codebook module BA; Nielsen & Chuang 1.4.1-1.4.4
- For next class: Codebook module GA; Nielsen & Chuang 6.1